

Workshop Activities

Activity 1 – Preattentive Attributes in a Data Visualization

Discuss each of the following statements about the use of preattentive attributes in a data visualization.

- Preattentive attributes can be used to draw the audience's attention to certain parts of a data visualization.
- Overuse of preattentive attributes can lead to clutter and can be distracting to the audience.

Activity 2 – Preattentive Attributes

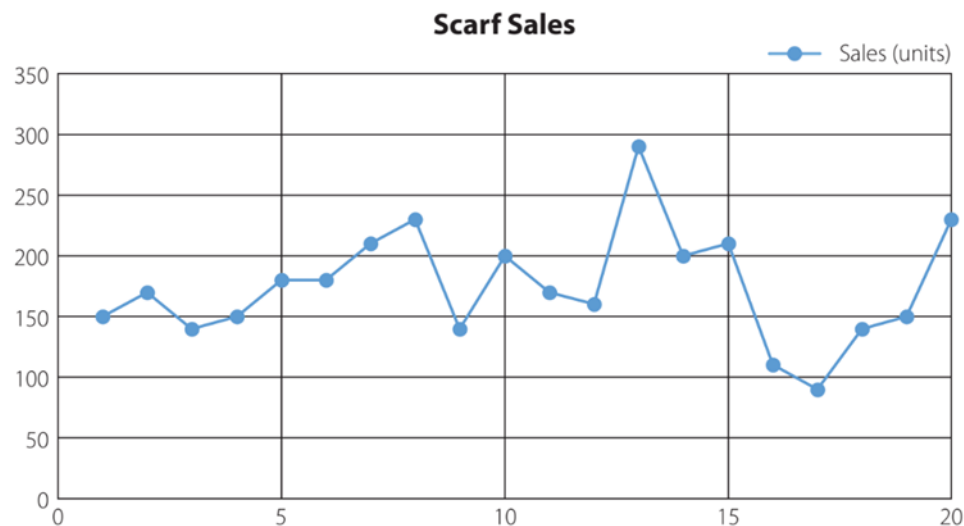
Consider the matrix of 160 single digits (16 rows x 10 columns) shown below.

7	3	4	1	3	4	5	6	4	0
3	0	6	9	0	4	5	8	6	3
2	7	2	2	9	9	4	5	2	1
2	2	4	5	2	0	9	2	0	4
2	4	0	7	6	9	3	0	0	4
7	7	8	9	2	6	7	2	4	7
6	1	3	3	2	1	4	4	9	0
3	6	6	2	7	5	5	2	5	4
1	1	4	0	6	3	4	0	5	1
3	7	5	2	7	5	7	7	3	9
3	3	8	6	9	5	5	3	6	4
7	6	0	3	0	9	9	0	2	9
4	6	9	4	8	2	6	5	8	3
9	3	9	2	2	8	4	3	9	8
5	8	8	2	9	1	2	4	8	5
1	7	4	0	1	1	9	9	5	8

We can aid in counting the number of 7s in the matrix using the preattentive attributes of hue and size. Regardless of the specific hue and size, discuss least three more ways you could use individual preattentive attributes to aid in counting the number of 7s in the matrix of single digits.

Activity 3 – Improving the Data-ink Ratio

Consider the Scarf Sales line chart shown below.

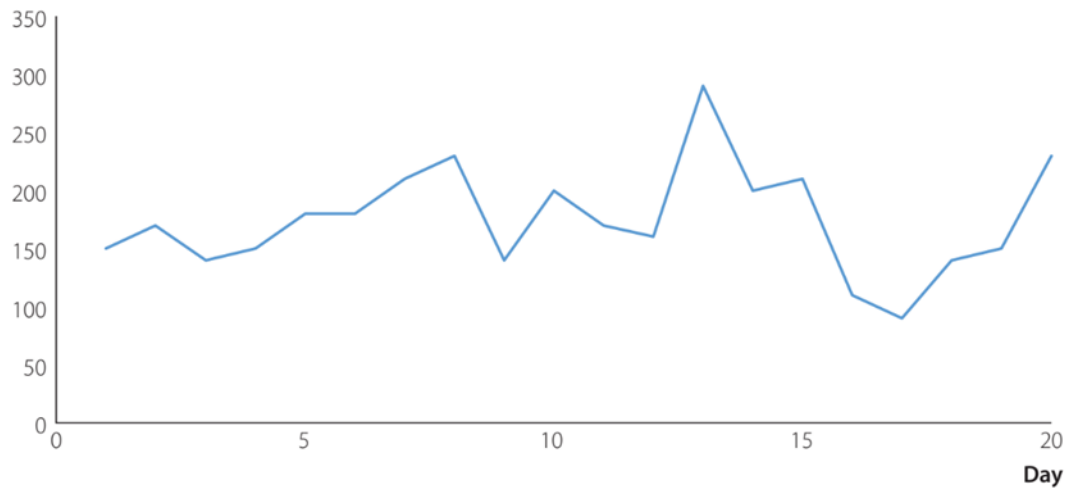


Default charts created in Excel often have a low and suboptimal data-ink ratio. You can build the default line chart from the *ScarfSalesChart* data file in Excel by selecting the A1:B21 cell range and selecting the ribbon's Insert tab. Then, in the Charts group, select the Insert Line or Area Chart icon and click on the top-left option under 2-D Line.

1. When building the default line chart, try increasing its data-ink ratio by removing any redundant and unnecessary feature to the data visualization. Do not add any features. List all the steps you implemented to increase the data-ink ratio.
2. Once you have removed all unnecessary and redundant features, it is now time to make the chart more readable by adding features so that it looks like the chart on the next page. List all the steps you implemented.

Scarf Sales

Sales (units)



Activity 4 – Creating a Scatter Chart using Excel

The next activity is to create a chart using data from the *Snow* data file.

	A	B	C	D
1	City	State	Average Low Temp	Average Snowfall
2	Atlanta	Georgia	53	2.9
3	Austin	Texas	59	0.6
4	Baltimore	Maryland	45	20.2
5	Birmingham	Alabama	53	1.6
6	Boston	Massachusetts	44	43.8
7	Buffalo	New York	40	94.7
8	Charlotte	North Carolina	49	4.3
9	Chicago	Illinois	41	36.7
10	Cincinnati	Ohio	43	11.2
11	Cleveland	Ohio	43	68.1
12	Columbus	Ohio	44	27.5
13	Dallas	Texas	57	1.5
14	Denver	Colorado	36	53.8
15	Detroit	Michigan	42	42.7
16	Hartford	Connecticut	40	40.5

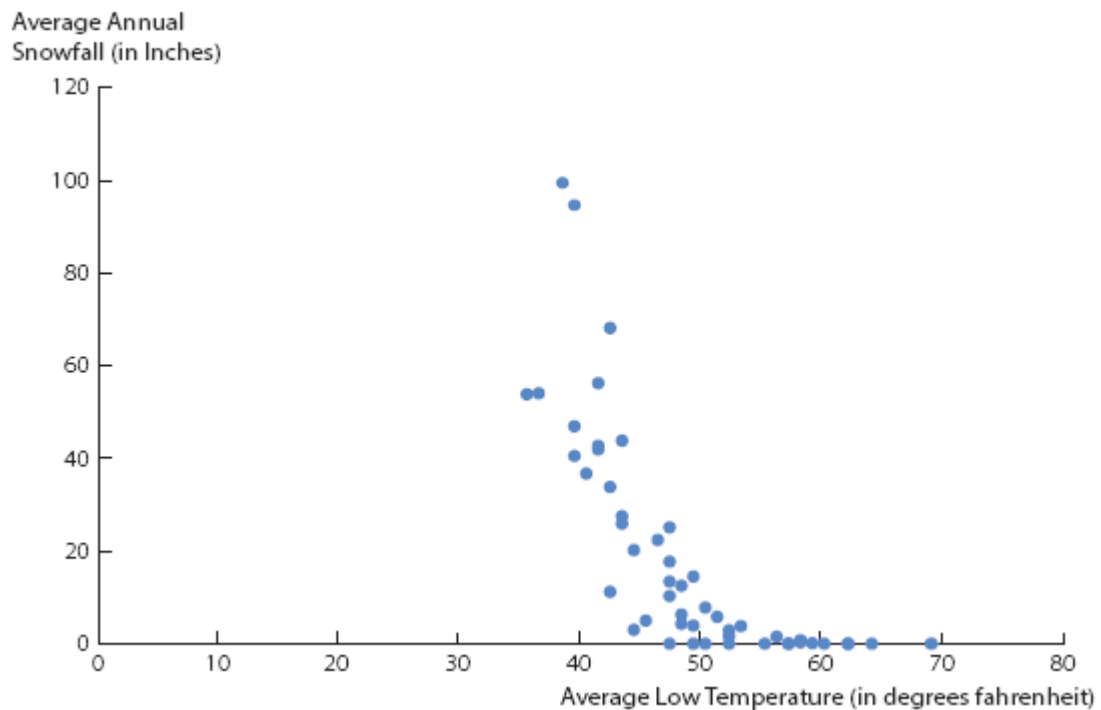
Open the *Snow* data file and complete the following steps:

- Step 1.
 - Select cells C1:D52
- Step 2.
 - Click the Insert tab on the Ribbon
- Step 3.
 - Click the Insert Scatter (X,Y) or Bubble Chart button in the Charts group
 - When the list chart subtypes appears, click the Scatter button

Then edit the chart to improve its appearance.

Your finished scatter chart should look similar to the following chart.

Average Snow Fall versus Average Low Temperature for 51 U.S. Cities



Each point on the chart represents a pair of numbers. In this case, it is a pair of measurements for each of the 51 cities. Write down the conclusions that you can draw from this chart.

Activity 5 – Creating a Bubble Chart using Excel

This activity creates a chart using data from the *AirportData* data file. The file *AirportData* contains data on a sample of 15 airports.

	A	B	C	D
1	Airport Code	TSA wait time (minutes)	Cheapest on-site parking cost (\$ per day)	Annual Enplanements (millions of passengers)
2	ATL	10.30	\$14.00	49.06
3	CLT	9.45	\$7.00	22.19
4	DEN	8.35	\$8.00	27.02
5	AUS	5.30	\$8.00	5.80
6	STL	4.80	\$7.00	6.25
7	SMF	7.30	\$10.00	4.60
8	RDU	6.75	\$6.50	4.80
9	EWR	9.90	\$18.00	18.80
10	SFO	7.25	\$18.00	24.00
11	LAX	6.40	\$12.00	36.10
12	SLC	4.50	\$10.00	10.80
13	SAN	8.80	\$15.00	9.70
14	MSY	4.60	\$16.00	5.30
15	IAD	3.0	\$10.00	10.70
16	BNA	2.50	\$12.00	5.60

Open the *AirportData* data file and complete the following steps:

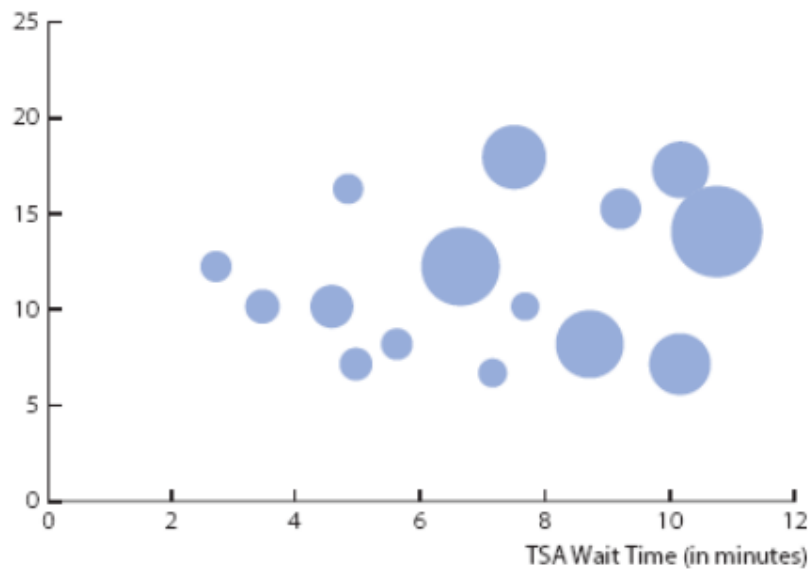
- Step 1.
 - Select cells B1:D16
- Step 2.
 - Click the Insert tab on the Ribbon
- Step 3.
 - Click the Insert Scatter (X,Y) or Bubble Chart button in the Charts group
 - When the list of chart subtypes appears, click the Bubble button.

Then edit the chart to improve its appearance.

Your finished bubble chart should look similar to the following chart.

Airport Passengers, TSA Wait Time and Cost of Parking

Cheapest Parking
Rate (in \$)



Write down the conclusions that you can draw from this chart.

In a bubble chart, you might wish to change which variables correspond to the x (horizontal) values, the y (vertical) values, and the bubble sizes. Once the chart has been created in Excel, the following steps can be used to change these assignments.

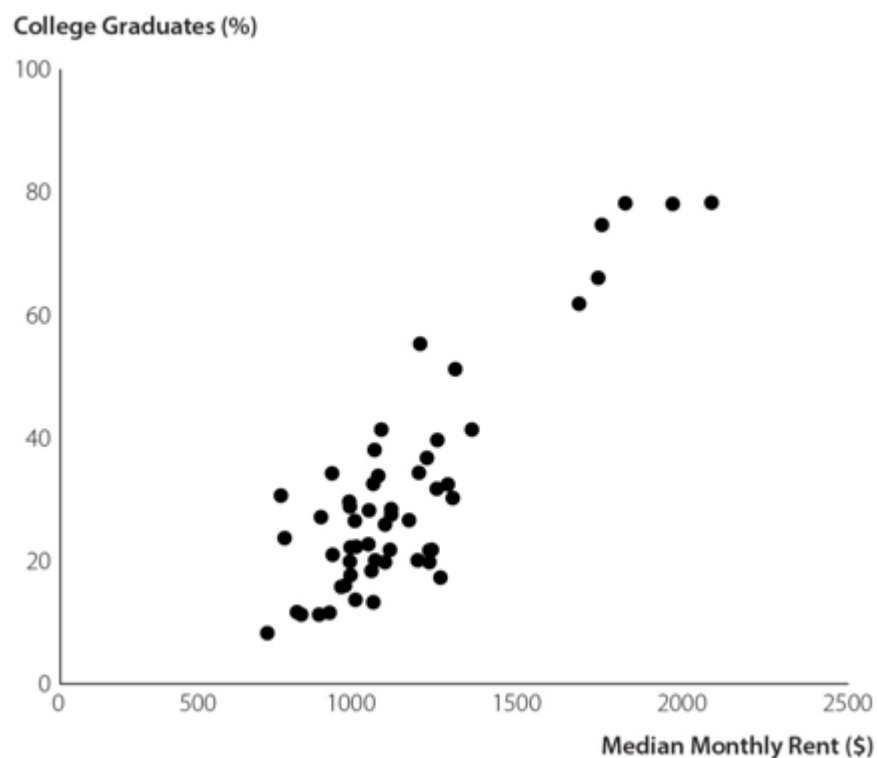
- Step 1.
 - Right-click any bubble
- Step 2.
 - Click Select Data...
- Step 3.
 - When the Select Data Source dialog box appears, click the Edit button under Legend Entries (Series)
- Step 4.
 - Enter the location of the data you want to correspond to the horizontal values in the Series X values: box. Do not include column headers.
- Step 5.
 - Repeat Step 4 for Series Y Values: box and the Series bubble size: box
- Step 6.
 - Click OK

Compare the two charts and describe whether these changes have improved the readability and understanding that can be gained from the data.

Activity 6 – Interpreting Scatter Charts

Percentage of College Graduates and Median Monthly Rent in New York City.

The following scatter chart shows the relationship between the percentage of college graduates and the median monthly rent in different sub-boroughs (a description of an area within the city) of New York City. Each point in the scatter chart represents a different sub-borough in New York City.



1. Which preattentive attribute is used in this scatter chart?
2. What can you say about the relationship between the percentage of college graduates and the median monthly rent in sub-boroughs of New York City?